

Analytical and Numerical Investigation of Two-Dimensional Projectile Motion

Physics Mechanics 101 Course Staff

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Abstract

This paper presents an analytical and numerical investigation of two-dimensional projectile motion under the idealized assumptions of constant gravitational acceleration and negligible air resistance. Beginning from Newton's second law, we derive the governing kinematic equations for horizontal and vertical motion and show that they decouple into two independent one-dimensional problems. Closed-form expressions for the time of flight, maximum height, and horizontal range are obtained and evaluated numerically for a launch speed of $v_0 = 25$ m/s across a range of launch angles. Trajectories and the range-angle relationship are visualized, confirming the well-known result that range is maximized at a launch angle of 45° on level terrain.

1 Introduction and Kinematic Foundations

Projectile motion is the archetypal example of two-dimensional (2D) kinematics: an object is launched with initial speed v_0 at an angle θ above the horizontal and subsequently moves under the influence of gravity alone. The central simplification that makes this problem analytically tractable is the *independence of the horizontal and vertical components of motion*. Because gravitational acceleration acts exclusively along the vertical axis, the initial velocity vector may be decomposed into orthogonal components,

$$v_{x0} = v_0 \cos \theta, \quad v_{y0} = v_0 \sin \theta,$$

and each component evolves according to its own one-dimensional equations of motion: the horizontal component under zero acceleration, and the vertical component under constant acceleration $-g$. This decoupling allows the full two-dimensional trajectory to be reconstructed by solving two independent one-dimensional problems and combining the results parametrically in time.

2 Derivation of Governing Equations

Taking the launch point as the origin, with x measured horizontally and y measured vertically upward, the equations of motion under constant gravitational acceleration g are

$$x(t) = v_0 \cos \theta t, \tag{1}$$

$$y(t) = v_0 \sin \theta t - \frac{1}{2}gt^2. \tag{2}$$

The corresponding velocity components follow by differentiation with respect to time:

$$v_x(t) = v_0 \cos \theta, \tag{3}$$

$$v_y(t) = v_0 \sin \theta - gt. \tag{4}$$

Equation (3) shows that the horizontal velocity is constant throughout the flight, while Equation (4) shows that the vertical velocity decreases linearly, reaching zero at the apex of the trajectory. Setting $v_y(t) = 0$ gives the time to reach maximum height, $t_{\text{apex}} = v_0 \sin \theta / g$. Assuming the projectile lands at the same elevation from which it was launched ($y = 0$), the total time of flight is twice the time to apex:

$$T = \frac{2v_0 \sin \theta}{g}. \quad (5)$$

Substituting t_{apex} into Equation (2) yields the maximum height,

$$H = \frac{(v_0 \sin \theta)^2}{2g}. \quad (6)$$

Finally, substituting the total time of flight T into Equation (1) and applying the double-angle identity $2 \sin \theta \cos \theta = \sin(2\theta)$ gives the horizontal range,

$$R = \frac{v_0^2 \sin(2\theta)}{g}. \quad (7)$$

Equation (7) is maximized when $\sin(2\theta) = 1$, i.e. at $\theta = 45^\circ$, independent of the launch speed v_0 .

3 Trajectory and Parametric Analysis

Figure 1 shows trajectories computed from Equations (1)–(2) for a fixed launch speed of $v_0 = 25$ m/s across five representative launch angles. Each curve's apex, corresponding to the maximum height given by Equation (6), is marked with a filled circle. As θ increases from 15° toward 75° , the trajectories become progressively taller and narrower, trading horizontal range for vertical reach.

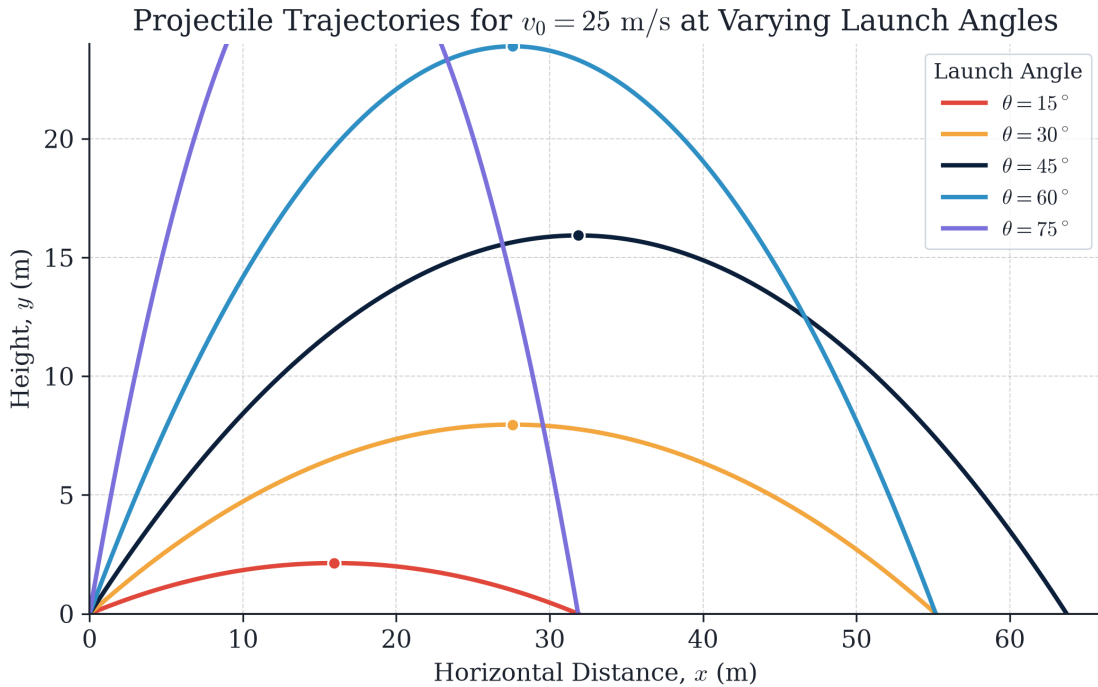


Figure 1: Projectile trajectories for $v_0 = 25$ m/s at launch angles $\theta \in \{15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ\}$. Filled circles mark the maximum height of each trajectory.

Figure 2 evaluates Equation (7) continuously over $\theta \in [0^\circ, 90^\circ]$, illustrating the symmetric, single-peaked relationship between range and launch angle. The range vanishes at both $\theta = 0^\circ$ (a purely horizontal launch that never gains altitude) and $\theta = 90^\circ$ (a purely vertical launch with no horizontal component), and is maximized at $\theta = 45^\circ$, as predicted analytically in Section 2.

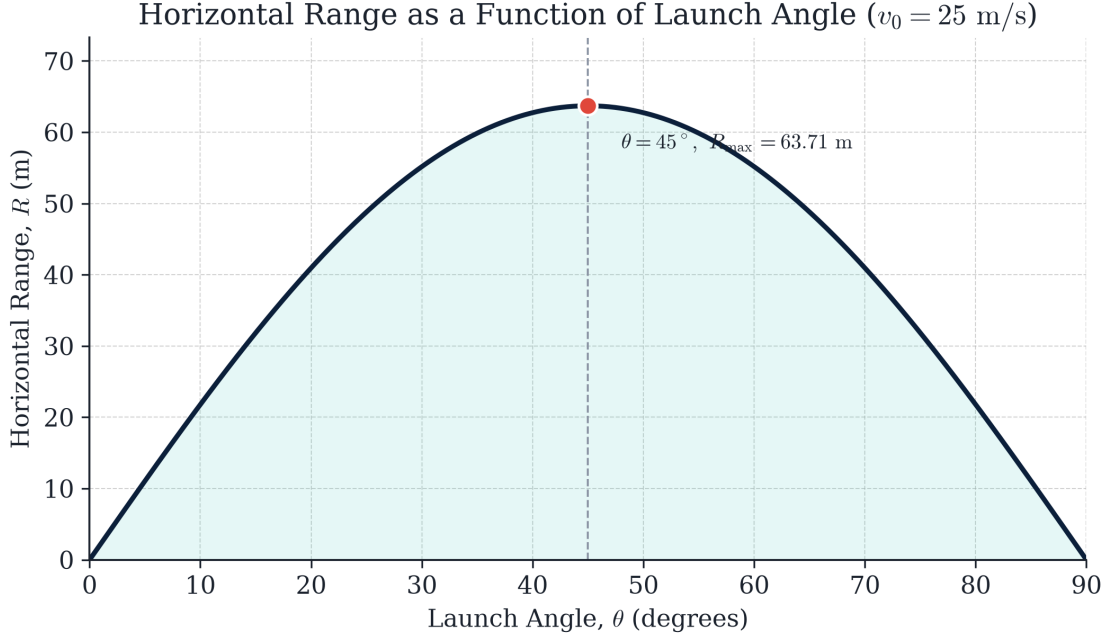


Figure 2: Horizontal range R as a function of launch angle θ for $v_0 = 25$ m/s, with the maximum range at $\theta = 45^\circ$ highlighted.

4 Quantitative Summary Table

Table 1 lists the time of flight T , maximum height H , and horizontal range R computed from Equations (5)–(7) for the same five launch angles shown in Figure 1, with $v_0 = 25$ m/s and $g = 9.81$ m/s² held fixed.

Table 1: Time of flight, maximum height, and range for $v_0 = 25$ m/s at five launch angles.

Launch Angle θ	Time of Flight T (s)	Max Height H (m)	Range R (m)
15°	1.319	2.134	31.855
30°	2.548	7.964	55.175
45°	3.604	15.928	63.710
60°	4.414	23.891	55.175
75°	4.923	29.721	31.855

5 Conclusion and Discussion

The analysis presented here confirms, both analytically and numerically, that idealized two-dimensional projectile motion under constant gravity decouples into independent horizontal and vertical one-dimensional problems. The derived closed-form expressions for time of flight, maximum height, and range (Equations (5)–(7)) accurately describe the trajectories in Figure 1 and the range–angle relationship in Figure 2, with

the maximum range occurring at $\theta = 45^\circ$ as expected from the $\sin(2\theta)$ dependence in Equation (7). This idealized model neglects air resistance, which in practice reduces range and introduces an asymmetric trajectory with an optimal launch angle typically below 45° ; extending this analysis to include quadratic drag is a natural next step and is treated in a companion numerical study.